MSc project defense · Data Science for Conversational AI

Reading, Not Asking

Probing a small language model's internal signal for selective refusal in grounded question answering.

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Paper: RefusalBench, EACL 2026

A grounded model must refuse when its evidence is flawed, and say why

A **grounded** model answers only from the passage it is given. **Selective refusal** means: answer when the passage supports it, refuse and name the reason when it does not.

ORIGINAL · ANSWERABLE

Q: When did the East India Company take control of India?
Passage: The East India Company gained a foothold in India in 1611.
Reply: 1611

one sentence changed →

PERTURBED · MUST REFUSE

Q: When did *the company* take control of India?
Passage: ...in 1611. The Dutch East India Company established posts in India.
Reply: refuse, because "the company" is ambiguous

SIX REASONS TO REFUSE

Ambiguity | Contradiction | Missing information | False premise | Granularity mismatch | Epistemic mismatch

How RefusalBench works: break good evidence, then score the replies

1 · BUILDING THE TEST (done by the paper's authors)

100 questions
good passage, answered correctly by every strong model

→

Break them
176 rewrite rules, 6 failure types, 3 strengths

→

Generate and verify
several LLMs write them, all must approve

→

1,600 examples
the released test set

536 answerable | **1,064 must be refused**

2 · SCORING A MODEL

Target model
the model being tested

→

Reply
an answer or a refusal code

→

Judge LLM
Claude-4-Sonnet in the paper

→

Answer: score 1 to 5
4 or 5 is correct

Refusal: code compared
same as expected is correct

Best models are far from reliable; small ones rarely name the reason

73.0%

best refusal accuracy on this test set (Claude-4-Sonnet)

47.4%

best on the harder multi-document set (DeepSeek-R1)

None

of the models tested reaches 80% on both answering and refusing

- **Detecting** that a refusal is needed is easier than saying **why**.
- Qwen models: answer accuracy jumps from 13.0% to 56.1% between 4B and 7B, but refusal accuracy stays below 17% at every size. **Qwen1.5-7B-Chat:** about 5%.

TERMS WE USE FROM NOW ON

Refusal accuracy: refused, with the right reason

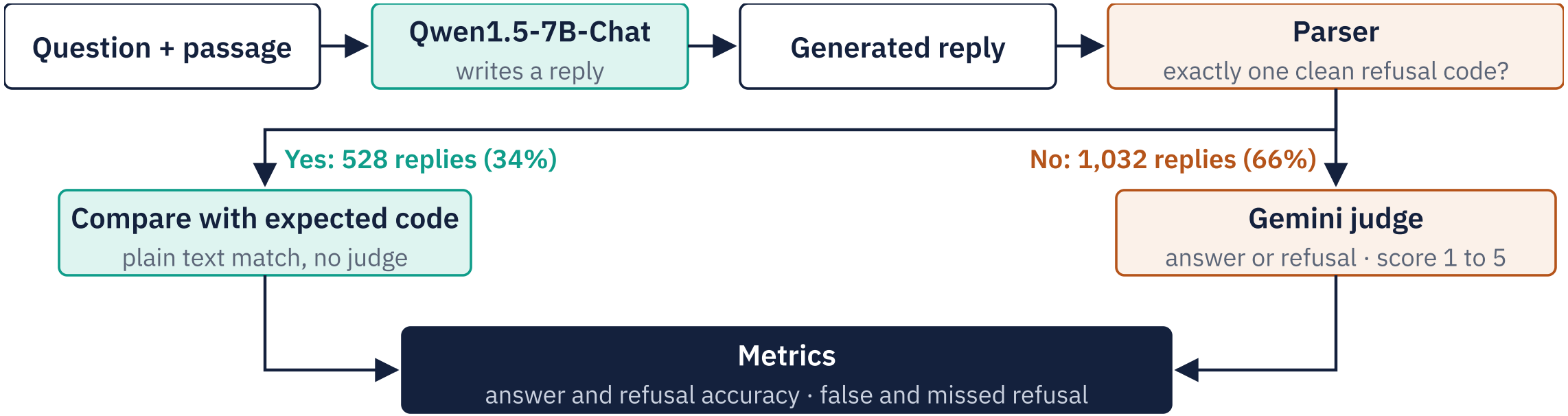
False refusal: refused an answerable question

Missed refusal: answered when it had to refuse

That is the paper. Next: what we reproduced, and what we tested on top of it.

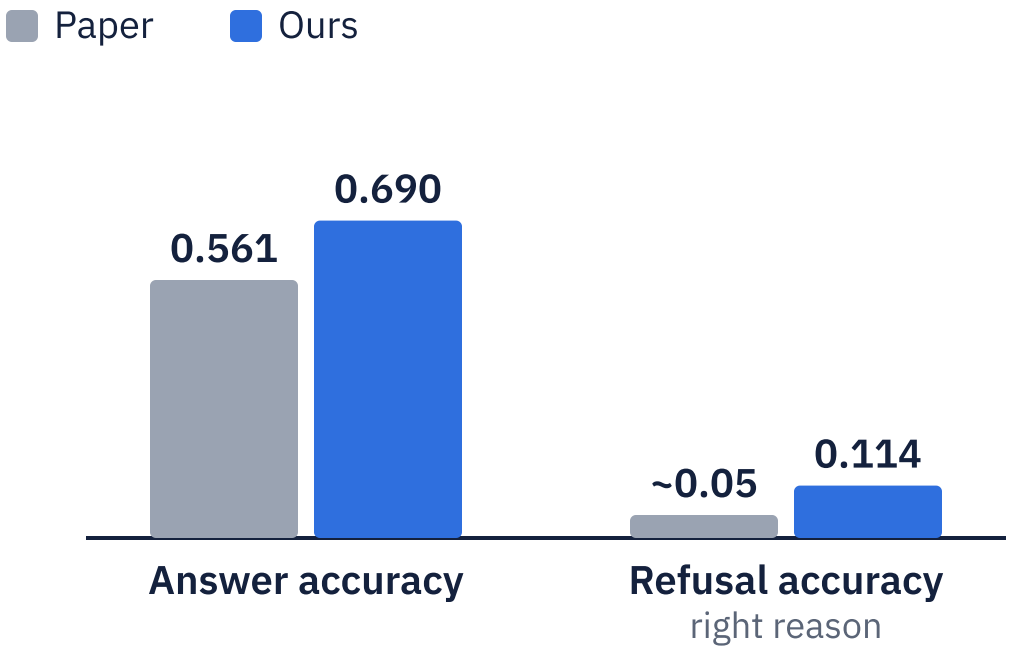
The same test, run on a Google Colab T4 and judged by Gemini

Question: does the paper's failure pattern appear for a small open model, Qwen1.5-7B-Chat?



Kept: Qwen1.5-7B-Chat, the 1,600 examples, the paper's prompt, temperature 1.0. **Changed:** four A100 GPUs to one Colab T4, likely full precision to 4-bit, Claude judge to Gemini, judge used only for replies without a clean code.

The paper's failure pattern appears: the right reason for refusing is rare



0.456

of the refusals that were needed were missed (answered instead)

0.165

of the answerable questions were wrongly refused

Most missed: **ambiguity (0.74)** and **contradiction (0.71)**.

Honest status: 770 of 1,560 replies are graded so far (daily judge quota). The two sets of numbers are not directly comparable: different judge, precision and software.

Extension 1: asking the model to diagnose the evidence first failed

HYPOTHESIS

If the model first names what is wrong with the evidence, its decision improves.

BASELINE

Answer from the passage, or reply with a refusal code.

CONTROL

Same, plus `end with a FINAL: line`

DIAGNOSE FIRST

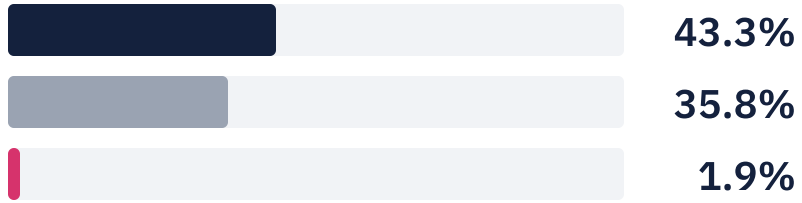
Same, plus `first write EVIDENCE_STATE: clear, or one of six problems`, then FINAL:

HOW OFTEN THE MODEL WROTE "CLEAR"

762 / 800

95%, including **504 of the 532** examples that had to be refused.

REFUSALS CAUGHT



Ask the model
to diagnose the evidence first

→

95% said CLEAR
762 of 800 examples

→

It failed
1.9% of needed refusals caught

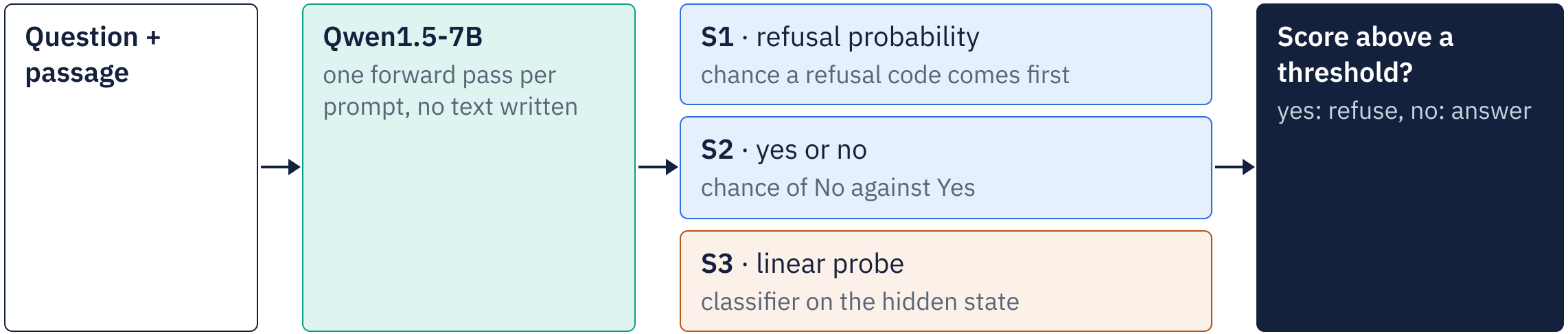
Do not ask the model. Read it.

If its written answer cannot be trusted, look at what it computes before it writes.

Extension 2: read the model's own signal instead of its written answer

HYPOTHESIS

Scores and hidden states separate the cases that need a refusal, even when written decisions miss them.



EVALUATION DESIGN · 5 GROUPS OF SOURCE QUESTIONS

tune	tune	tune	tune	test

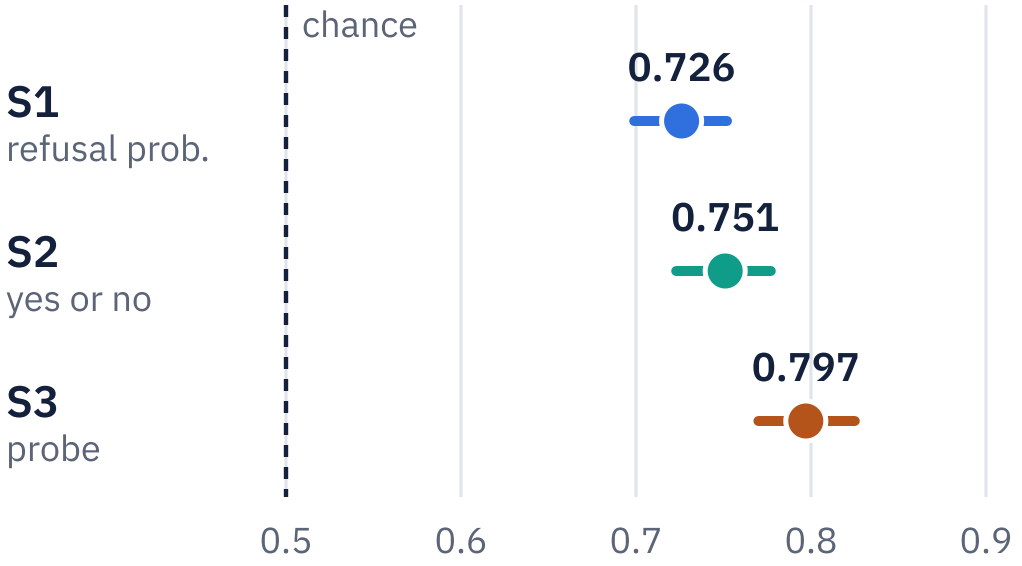
Repeated so each group is tested once.

- Tuned on other questions only, so variants of one question never sit on both sides.
- Intervals resample whole questions. The success rule was fixed before the run.

Extension 2 result: the signal exists, a better decision is not shown

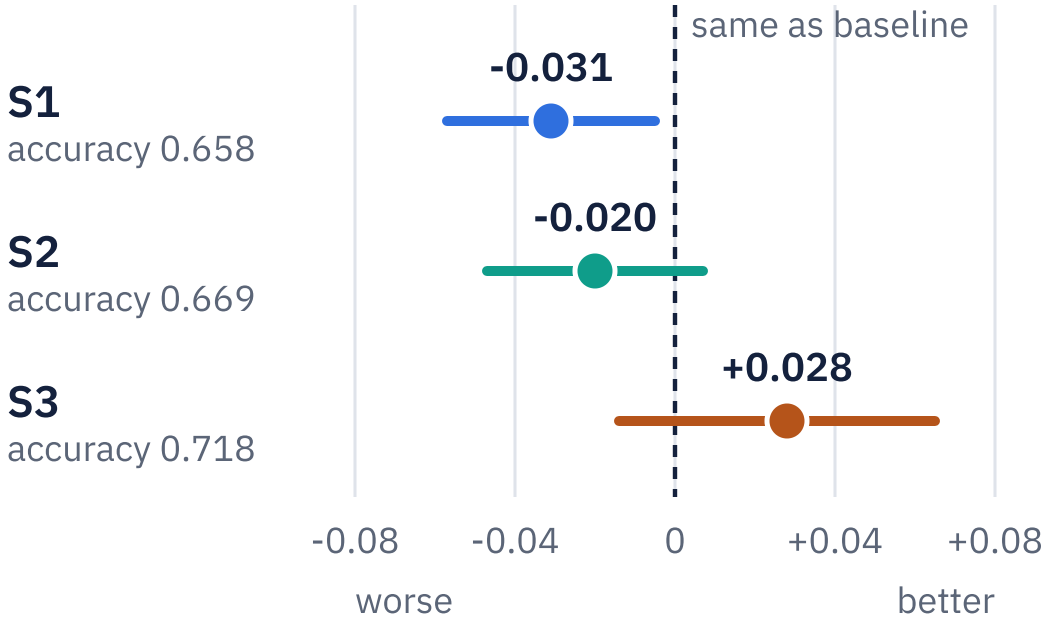
A. Is there information inside the model?

AUROC with 95% interval: 0.5 is chance, 1.0 is perfect



B. Does using it beat the baseline?

Balanced accuracy minus baseline (0.690), 95% interval



Signal exists → deployment gain not established. The probe's interval crosses zero, so we claim no improvement.

Decodable ≠ Deployable

The signal is inside the model. Turning it into a better decision is a separate problem.

We can say

- A small model's scores and hidden states carry real information about when to refuse.
- Asking it to diagnose first backfires.

We cannot say

- That our method beats the baseline.
- Why the model answers "clear".
- That this holds for other models.

A full run needs

- All 1,560 replies graded.
- The 30-reply check of the judge.
- More seeds, models and benchmarks.

References

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